

Potential of legume-based cropping systems for climate change adaptation and mitigation

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20.1 Introduction

The unprecedented increase in population exists as a great apprehension for the researchers and policy makers across the globe for ensuring safe and nutritional foods. The present world population is 7.82 billion, while the projected number in 2050 and 2100 will be 9.7, and 10.9 billion, respectively (UN, 2019). The world presently is passing through several stern challenges related to agriculture and the environment originated from overpopulation (Fig. 20.1). Kumar et al. (2020) also identified some crucial challenges like food and nutritional security, soil and land degradation, global warming, and climate change. To feed the ever-burgeoning world population, food production must be increased through high input-intensive agriculture from finite land resources (Rahman et al., 2020a). It is appraised that production of foods to be augmented by 60%–110% in 2050 compared with the amount in 2005–2007 (Alexandratos and Bruinsma, 2012; Pugh et al., 2016). Cereal-based intensive agriculture greatly influenced to upswing global food production, which conversely caused malnutrition in a large number of the world population (Jangir et al., 2017a). According to FSIN (2020) estimation, over 130 million people around the world are in acute diet uncertainty. Population of climate-vulnerable African and Southeast Asian countries grieved more from malnutrition. The long-term consequences of intensive agriculture are

degraded land and water resources, changed global climate, polluted environment, etc. which the people are fronting today. Intensive agriculture and deforestation further diminish biodiversity and turn to stimulants to frequent natural hazards. Forest clearing and biomass burning emit a considerable amount of greenhouse gases (GHGs) from the earth surface to the atmosphere across the globe which further contributes toward a changing climate. The outcomes of the total scenarios are a malnourished society, unsustainable agriculture, and ecological disruption in many corners of the world (Fig. 20.1).

Climate change is one of the most prevailing human-induced issues facing the planet because of heavy dependence on synthetic nitrogen (N) fertilizers, fossil fuel burning, industrialization, and modern transport systems that emit significant amounts of GHGs (Hussain et al., 2019). About 76% of the total emission is carbon dioxide (CO₂), while methane (CH₄) covers 16%, nitrous oxide (N₂O) 6%, and F-gases only 2% (IPCC, 2013). The share of agriculture to GHGs emissions has been reported to be 24% (IPCC, 2013). Synthetic N fertilizer released 241.27 Gg (Giga gram) of N₂O in 1961, while it jumped to 2266.11 Gg in 2019 (FAOSTAT, 2021). Emission of N₂O also occurs because of different organic fertilizers applications including manure, traditional tillage (TT) practices, the aerobic process of composting, etc. (Oenema et al., 2005). It was reported that wastes originated from different animals

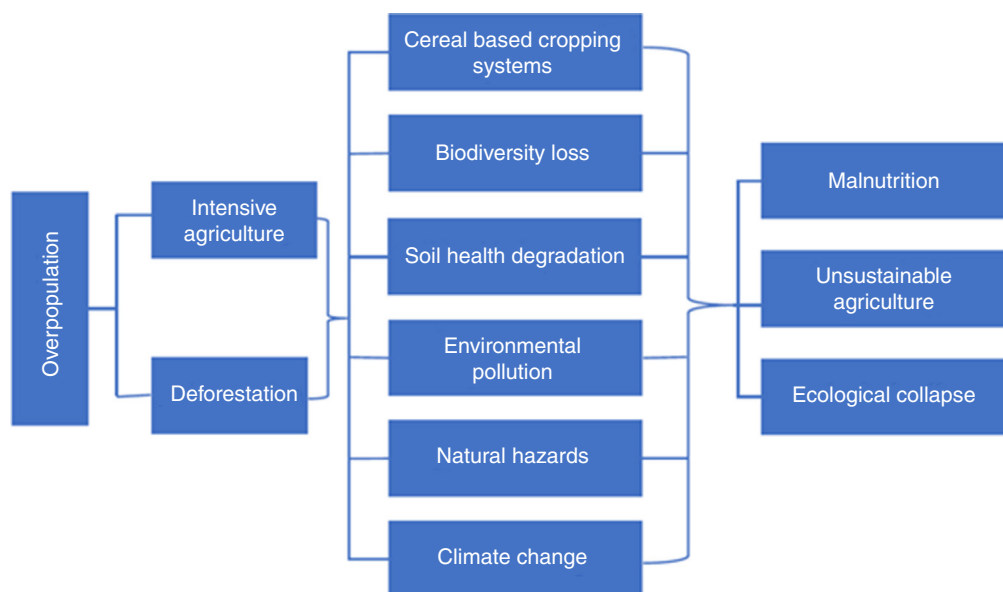


FIG. 20.1 Present global challenges related to agriculture and the environment.

contributed about 30%–50% of N_2O to the worldwide emission from agriculture (Oenema et al., 2005; Davidson, 2009). The emission of N_2O from soil to the atmosphere is a serious concern as it is a potent GHG and on a 100-year time scale, the global warming potential (GWP) of N_2O is 298 compared to 1 unit of CO_2 .

The deliberate management of agriculture and forest with diversified crops and tree species can be operative tools for climate change mitigation, ecological veracity, agricultural, and environmental sustainability. It has been reported that legumes can reduce the emission of GHGs by 5–7 times than other crops and increase annual carbon (C) sequestration in soils by $21.8 \text{ g C kg}^{-1} \text{ soil}$ (Stagnari et al., 2017). The inclusion of legume crops and trees in agriculture and agroforestry might ensure to achieve the sustainable development goals (SDGs) of the United Nations (UN), mainly to end hunger, achieve food security and improved nutrition and promote sustainable agriculture (SDG 2) and to ensure healthy lives and promote well-being for all at all ages (SDG 3) by 2030. Legumes through a symbiotic relationship with bacteria fix atmospheric dinitrogen (N_2) which is known as biological nitrogen fixation (BNF) (Rahman et al., 2020b). Meena and Lal (2018) stated that the land-based annual global supply of reactive N through BNF is 110 Tg (tera grams), while the industrial production by Haber–Bosch process is about 100 Tg, which signifies the status of legumes in the environment. The legumes maintain and improve soil fertility, reduce N fertilizer application, increase crop yields, sequester soil C, and thereby reduce CO_2 and N_2O emissions, and ultimately contribute to mitigating global warming and climate change (Meena and Lal, 2018). Effects of legumes on soil and environment and climate change mitigation are portrayed in

Fig. 20.2. Even though legumes can perform strategic and noble roles in safeguarding human beings and the environment, however, their importance is ignored in many cases or paid little attention. Therefore, this chapter attempted to provide scientific knowledge on legume-based diversified cropping systems including agroforestry and aesthetics. It was hypothesized that the cultivation of legumes may act as a one-stop solution for a healthy soil capable of producing healthy crops for ensuring a healthy generation in a healthy environment. The objectives of the chapter were to collect and collate information on the potential of legumes for improving soil health, reducing GHG emission, confirming nutritional security, identifying barriers to legume cultivation, and taking required policy action for removing barriers.

20.2 Climate change and its impacts: present status and future projection

Climate change is now a reality. There is circumstantial evidence of climate change like loss of biodiversity, increase in storms, tropical cyclones, floods, droughts, extreme cold waves or extremely high temperature, etc. (Steffen, 2006). The projected global warming because of GHGs, especially CO_2 will result in temperature rise between $1.8\text{--}4^\circ\text{C}$ by 2100 and sea-level rise between 180–590 mm (IPCC, 2007). Lindsey and Dahlman (2021) reported that the global surface temperature has risen by 0.08°C per decade since 1880, while the decadal rate was 0.18°C over the past 40 years since 1981. They also observed that the average surface temperature across land and ocean during 2020 was 0.98°C higher than the 20th-century average of 13.9°C . Research findings revealed that an increase in CO_2

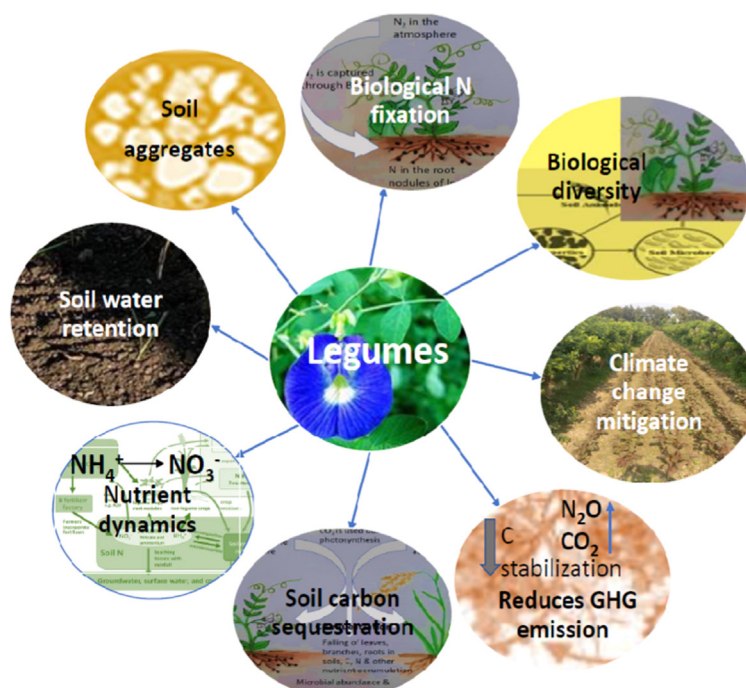


FIG. 20.2 Diversified effects of legumes on soil and environment toward climate change mitigation and agricultural sustainability.

concentration increases the photosynthetic efficiency of rice (*Oryza sativa* L.), wheat (*Triticum aestivum* L.), etc., and, thereby increases their yields (Parry et al., 2011; Kant et al., 2012). In contrast, the rise of temperature masks the effect of elevated CO_2 and reduces crop production, which is a great dilemma in agriculture. Climate change caused erratic rainfall which results in higher water requirements for most of the crops because of enhanced evaporation and transpiration (Tewari, 2012). From the end of 1900, the earth's atmospheric temperature has increased by 0.8°C , and at the same time period, the Arctic temperature has increased by 2 to 3°C (Overland et al., 2014). The anticipated impacts of temperature rise in the Arctic are ice melt and subsequent sea-level rise, loss of wildlife and biodiversity, and higher emission of GHGs.

Global warming and climate change and their resultant effects on different sectors of agriculture and the environment are obvious (Fig. 20.3). Because of climate change and the use of imbalanced fertilizers solely depending on inorganic sources, it is estimated that rice and wheat productivity in Bangladesh in 2050 will be declined by 8, and 32%, respectively, against a base year yield of 1990 (MoEF, 2008). Surplus N may augment microbial decomposition of organic matter (OM) which may decrease C content in soil and increase CO_2 emission (Jangir et al., 2019). Misuse of N enhances its loss as nitrate (NO_3^-) leaching, ammonia (NH_3) volatilization, and denitrification as N_2O and nitric oxide (NO). Therefore, crop cultivation in climate-vulnerable countries across the world is a great challenge.

A large drop in attainable yields of major cereals is projected across the globe by 2050 (Pugh et al., 2016). The atmospheric CO_2 concentration in 1950 was 310 ppm (parts per million) (Steffen et al., 2007), which has increased to 417.5 ppm in July 2021 (GML, 2021). The annual rate of CO_2 increment during this time period was roughly 1.51 ppm. The present concentration of atmospheric CO_2 is higher than recorded in the past 800,000 years (Lindsey, 2020). Biswas et al. (2019a) reported that with the increase of temperature by 1 – 3°C and CO_2 concentration from 380 to 550 ppm, grain yield of different varieties of wheat developed by the Bangladesh Agricultural Research Institute (BARI) was significantly decreased. In India, there is a possibility of crop loss of 10%–40% because of the projected temperature rise in 2100 (Tewari, 2012). It was also reported by Tewari (2012) that for each 1°C temperature rise in the growing season; rice grain yields are likely to be declined by 10% and rapeseed mustard by 3%–7%. On the other hand, temperature rise in the three Southeast Asian countries, namely, Indonesia, the Philippines, and Thailand contributed to increasing rice production linearly from 1960 to 2015 (Cortes and Otadoy, 2020). Climate change impacts on crop yields in European agriculture might be variable depending on crop types, and changes in temperature and CO_2 emission. Supit et al. (2012) projected that yield of winter wheat may increase until 2090 with the increase of CO_2 provided that there is a sufficient amount of rainfall during the growing season, while yield may decrease after 2050 if CO_2 increase is less and temperature rises is high.

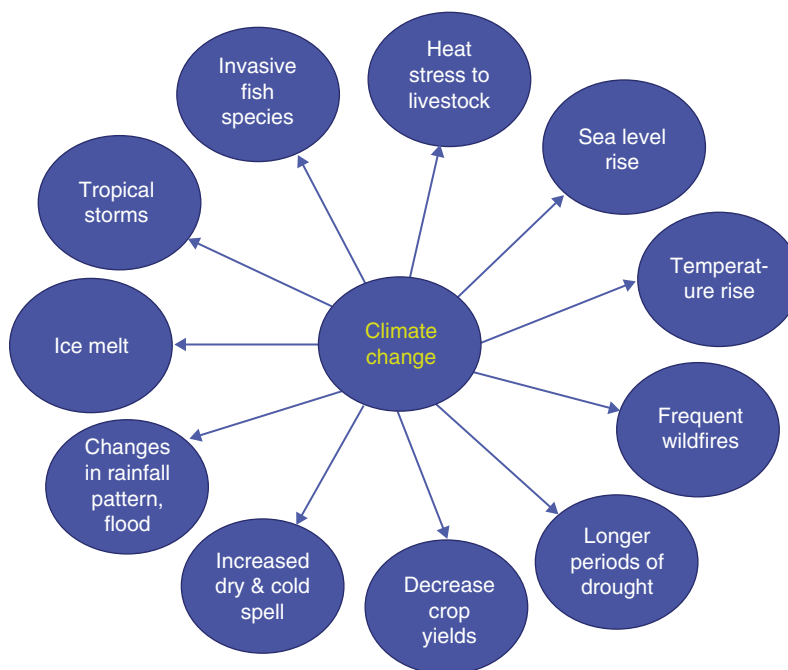


FIG. 20.3 Impacts of changing climate on different sectors of agriculture and the environment.

They also anticipated the yield reductions of spring crops like potato (*Solanum tuberosum* L.), maize (*Zea mays* L.), and sugar beet (*Beta vulgaris*), while at the termination of the 21st-century yield decline is likely in entire Europe if temperature rise is continued.

20.3 Climate change adaptation and mitigation

Adaptation and mitigation can significantly lessen the deleterious waves of climate change to crop production and the environment. Therefore, suitable strategies of climate change adaptation and mitigation need to be identified and practiced. A series of best management practices (BMPs) have been suggested to reduce the C emission from agriculture (Oenema et al., 2005). Diversified cropping systems, crop residue retention, conservation tillage (CT), sowing and transplanting time adjustment, water management, mulching, cover crops, etc., are known as BMPs. Such a set of BMPs along with the optimum and well-adjusted application of organic and inorganic fertilizers, etc., are believed to offer a higher capacity to raise C levels in soils. Adoption of BMPs may bring changes in soil organic and inorganic C levels, increase C stock and alter soil microclimate (Lorenz and Lal, 2018). Conservation and reduced tillage (RT) hold the ability to accelerate C stock and reduce the oxidation of C to CO₂. Thus, some of such practices might have the adaptive capacity, while others help to mitigate the negative consequences of global warming and climate change (Table 20.1).

To cope with the changing climate and unstable environments, adjustment in production technology is expected for climate-smart cropping systems. Abiotic stress tolerant crop, short duration, and low input crop varieties, and suitable crop calendar based on temperature and photoperiods have been considered as a long-term adaptation approach to climate change (Debaeke et al., 2017). The vulnerability of climate change can be addressed through introducing short-term adaptation strategies like shifting sowing date, adopting diversified cropping patterns including legumes, yield goal-based nutrients applications, integrated nutrient management, introducing crop demand-based irrigation facility and resource conservation strategies (Zhao et al., 2015; Debaeke et al., 2017; Kumar et al., 2022).

Like Bangladesh, there are other climate vulnerable countries especially India, Pakistan, and some areas of Africa are of great concern for achieving food security. Many areas of these countries are gradually going to be desert with a prolonged drought. Drought has become a great threat in this region, where crop loss ranges 20%–60% (Banglapedia, 2006; Bharadwaj et al., 2020). Legume-based cropping patterns could be needed for improving soil health, quality, and fertility under such climate change conditions (Fig. 20.2 and Table 20.1). It has been reported that the GWP of legume based cropping systems is much lower than that of rice-based cropping systems. The GWP (CO₂ eq.; carbon dioxide equivalent) of nonlegume systems rice-rice-fallow and mustard (*Brassica nigra*)-rice-rice were 8432, and 9131 kg ha⁻¹, respectively, while legume-based wheat-mung bean (*Vigna radiata*

TABLE 20.1 Soil and crop management practices considered for climate change mitigation and adaptation.

Mitigation/adaptation strategies	Impacts on soil and environment
Mitigation	
Zero till/minimum or reduced tillage	Increase C sequestration, reduce GHGs emission
Retention of crop residues and organic fertilizers application	Increase C sequestration, reduce N fertilizer application and thereby reduce N ₂ O & CO ₂ emission, improve soil health
Anaerobic composting of crop residues, manures and different organic materials	Reduce CO ₂ emission
Fertilizer deep placement and band application	Decrease N ₂ O emission, and increase nutrient use efficiency (NUE)
Afforestation and reforestation	Reduce deforestation and increase C sequestration
Legume-based agroforestry	Increase biomass & C sequestration, reduce CO ₂ emission, ensure biodiversity, improve soil health
Alley cropping system	increase C sequestration, reduce N fertilizer application and thereby reduce N ₂ O & CO ₂ emission, ensure biodiversity, improve soil health and crop yields
Crop rotation	Increase NUE, ensure biodiversity, improve soil health
Alternate wetting & drying	Reduce CH ₄ emission, increase NUE. Slightly increase N ₂ O emission.
Restoration of degraded lands	Increases C sequestration
Adaptation	
Adjustment of planting dates	Avoid adverse production environments like drought, flood etc.
Relay cropping	Efficient utilization of soil moisture, avoid drought
Genetic engineering and biotechnology	Improve stress-tolerant of crop varieties
Floating agriculture	Climate-smart technology in the coastal areas and long-term submerged soils, ensure nutritional security
Raised bed cropping	Increase NUE, reduce N ₂ O emission
Mulching	Conserve soil moisture and nutrient loss
Terracing/contour farming	Conserve soil, reduce nutrient loss

L. Wilczek)-rice and grass pea (*Lathyrus sativus* L.)-rice-rice systems were 3109 and 3643 kg ha⁻¹, respectively (Biswas et al., 2019b). It was stated that the addition of pigeon pea (*Cajanus cajan* L. Millsp.) in a cropping pattern increased total soil N content (100 mg kg⁻¹ soils) which was 4 times higher compared to without legume (25 mg kg⁻¹ soils) in the crop rotation (Kumar et al., 2020). Insect-pest infestation increasing with changing the climate. But appropriate plans for adaptive research with diverse crop rotations including legumes could be lessened the incidence of insect pests, weeds, and diseases infestation which successively increase 15%–25% rice production (Dahiya et al., 2017; Kumar et al., 2020).

Agricultural land is decreasing globally because of industrial growth, new settlements, constructions of roads, etc., and therefore, required food production is a great challenge. Rice is the staple food of South Asia and food

security depends on its ample supply and year-round accessibility. Besides, the repeated incidence of distressing flash floods, droughts, salinity, storms, etc., has reduced agricultural growth in South Asia which caused a distraction to the supply of foods. South Asia comprises the sub-Himalayan eight countries with less than 5% of the world's land mass and 14% of the global arable land. Per capita arable land of the region in 1961 was 0.351 ha, while it decreased to 0.114 ha in 2018 (World Bank, 2021). However, the region supplies food and supports the wellbeing to about 45% of Asia's population and 25% of the world's population (INMS, 2021). Except for Australia, it appeared from the base year of 1961 that the arable land in many countries sharply declined until 2018 as depicted in Fig. 20.4 (FAOSTAT, 2021). Among the selected countries, per capita, arable land was the highest in Australia (1.24 ha), while the lowest was in Malaysia (0.03 ha). Legume-based

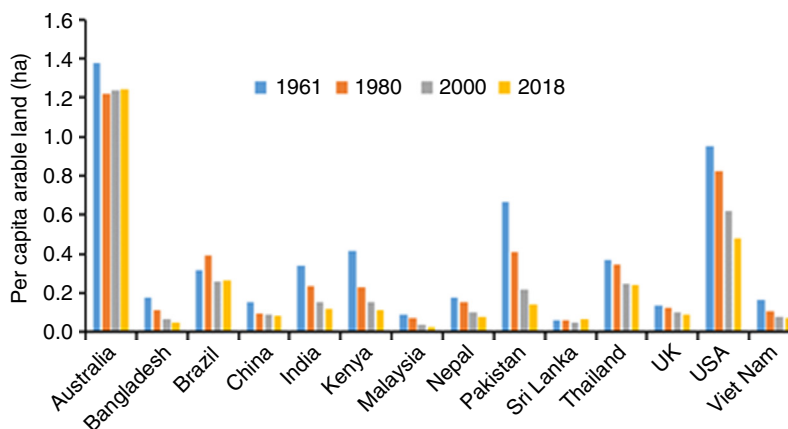


FIG. 20.4 Per capita arable land in different countries of the world (data source: FAOSTAT, 2021).

cropping systems might ensure higher production from the limited land resource.

There is a big question remaining for the poor farmers of the least developed and developing countries to maintain soil fertility by adding soil OM (Jakhar et al., 2017). Soil OM is declining day by day for intensive cropping in the nonagroforestry system (Rahman et al., 2016). While in agroforestry system, soil richness increases because of the application of a huge amount of organic inputs and N fixation which mineralize and release nutrients, help to develop soil structure and reduce soil erosion (Pinho et al., 2012; Lorenz and Lal, 2014). Agroforestry systems can either increase or at least maintain the OM contents in soil (Jose, 2009; Pinho et al., 2012). The system can increase NUE through reshuffling the soil nutrients by deep-rooted trees and reducing N fertilizer because of BNF by trees like gum Arabic tree (*Acacia nilotica*).

20.4 Legumes: a vital component of climate-smart cropping system

The cropping systems concept generally refers to crops, annual crop sequences, and management techniques for agricultural land use. Conventional cropping systems like agronomy, horticulture, forestry, and other types of land uses contribute approximately 24% of global GHGs emissions (Smith et al., 2013). Carbon sequestration through sustainable cropping system management and green energy production for substituting fossils fuel might reduce N_2O , CH_4 , and CO_2 emissions (Debaeke et al., 2017; Roy et al., 2021; Meena et al., 2022). Leguminous trees and crops belong to the family Fabaceae are the third largest family of flowering plants consists of 732 genera and 19320 species that can grow in a multiple cropping system (Borges et al., 2013). Legume-based ecosystems can afford multiple sustainability-related services. However, the dynamic and diverse potentiality of legumes in climate-smart cropping systems of future agriculture needs to be explored.

20.4.1 Field crop system

Immediately after cereals, globally legumes are considered as a highly consumed crop as food, feed, and biofuel (Stagnari et al., 2017). Such crops can be included in the climate-smart crop cultivation systems to intensify crop multiplicity, cut production inputs, and promote conservation agriculture. Climate and environmental change might create a harsh crop production situation which can be a great threat to sustainable agriculture and food security (Dahiya et al., 2018). Legumes can be used for phytoremediation of harsh soil environments due to their high potentiality of biomass production, and have been identified as a potential climate change coping strategy within smallholder farming systems. Grain legumes in the crop production scheme can considerably increase the yield of succeeding crops of the rotation and to a lesser extent through intercropping. Angus et al. (2015) found 30% more average yield in legume-wheat rotations than that of wheat-wheat monocropping. Gao et al. (2010) witnessed that a total of 65, and 71% yield was increased in maize intercropped with soybean (*Glycine max* L. Merr.) as compared with both crops grown as a monoculture. Legume fellow crops generally have attained more N uptake, namely, pea (*Pisum sativum* L.) and narrow-leaved lupin (*Lupinus angustifolius*) followed by cereals have increased N uptake by 23%–59% (Jensen et al., 2004). Yields of rapeseed (*Brassica napus* subsp. *napus*) and potatoes might be increased if precrops are legumes. In intercropped cereals, planting legumes may contribute up to 15% of the N required for the cereal crop, thus biomass production increases and synthetic N-fertilizer use reduces (Li et al., 2009). When legumes are grown as winter crops in rice-bean and rice-faba bean (*Vicia* spp.) system, the N concentration in rice residue is amplified by 9.7%–20.5%, while its content in soil ranges from 0.187% to 0.193% (Yu et al., 2014). Legume-based intercropping might be accelerating the linkage of mycorrhiza with the roots system of legumes which improves P availability (Ren et al., 2013).

Moreover, legume roots are capable to modify the soil pH and P availability (Li et al., 2013). Legumes have versatile benefits in sustainable crop production systems. But the use of legumes in the cropping systems is yet a great concern because of their low and unstable yields, and inadequate policy support (Bues et al., 2013).

Increase C stock and C sequestration in soil and biomass has been considered as a promising mitigation approach for lessening climate change impact. Legume-based cover crops in the arable cropping system, legume-based grassland and agroforestry helps to develop soil structure, promote positive C balance, and reduce GHG production (Autret et al., 2016; Lal, 2017). Legumes may improve soil health, and reduce input requirements, and thus have a potential role in mitigation and adaptation of climate change. Legumes can reduce GHGs emissions by lessening the CO₂ and N₂O fluxes, reducing fertilizer use, enhancing C sequestration, and lowering burning fossil fuels in the agricultural production systems. The inclusion of N-fixing leguminous plants in rotations either as a key crop or cover crop becomes a vital and profitable tool for reducing N₂O emissions. Legumes not only efficiently control N₂O emission but also renovate N balance and thus helps in minimizing the N rate for the next crop. Legume-based cropping systems provide an agricultural eco-friendly N management system as such system ensures soil health, water quality, air quality and preserves biodiversity (Debaeke et al., 2017; Sánchez-Navarro et al., 2020).

Field crop agriculture emits GHGs and consumes CO₂ that affect climate change. Field crops require a huge amount of urea N, which hydrolyzes rapidly by urease enzymes to NH₃ and CO₂ (Jenkinson, 2001). Research findings revealed that from 1 kg of industrial N production releases 2.6–3.7 kg of CO₂, and thus, global annual production of N releases 300 Tg of CO₂ into the atmosphere (IFA, 2010). N fertilizer saving because of BNF through legumes in rotation annually reduces almost 277 kg CO₂ ha⁻¹ (Jensen et al., 2012). Legume residues help to form stable organic carbon (OC) fractions and reduce CO₂ emissions to a certain extent. It is learned that N₂O emission contributes to about 5%–6% of total GHG emissions, but its GWP is almost 300 times that of CO₂ (Crutzen et al., 2016). Among different agricultural activities, N fertilizer greatly contributes to N₂O emission, and it is foreseen to escalate by 30%–60% in the coming 20 years (Reay et al., 2012). It is calculated that roughly 1.57% N₂O is emitted from N fertilizer application to crop fields (Jensen et al., 2012). Compared to other crops, legumes can reduce GHG emissions. Jeuffroy et al. (2013) found that N₂O emissions by peas, winter wheat, and rapeseed were 69, 368 and 534 kg ha⁻¹, respectively.

The C sequestration in the soil increases stable C form like humus which ultimately lessening the CO₂ emission (Christopher and Lal, 2007). Different cationic forms of soil nutrients like N, potassium (K), calcium (Ca), etc., and

anionic forms of N, phosphorus (P), and sulfur (S) as well are bound to C in humus. It is noted that every ton of organic C accumulated in soil contributed approximately 80 kg of N (Kirkby et al., 2011). It has been reported that the conversion of grassland to cropland has caused an annual decline of soil C stocks by 25%–43% (Soussana et al., 2004). Soil OC, N, P, K, S, etc., and humus content increase in soil under legume-based cropping systems (Jensen et al., 2012). Legumes can increase C content by providing more biomass, organic C, and N in soils (Garrigues et al., 2012). The application of soybean residues can increase SOC by 38.5% (Bichel et al., 2016). Rochester et al. (2001) found from a 10-years study that the legume-based crop rotation restored residue-C by more than 49%, residue-N by more than 133%.

20.4.2 Agroforestry system

The agroforestry system appears as a valuable and cost-effective technology that can address climate change issues smartly. The system encompasses a large number of diverse land-use methods at the interface of agriculture, livestock, and forestry. Legume agroforestry systems through the addition and decomposition of litter and biomass can increase total and available N, and also other nutrients in the soil. A long-term study conducted by Sileshi et al. (2012) in Malawi and Zambia, publicized that the yield of intercropped gliricidia (*Gliricidia sepium*)-maize was 42% higher than that of unfertilized fields, which was alike to maize yield obtained from the fertilized plot that received 92 kg N ha⁻¹. In addition, compared with the fertilized field, the yields of gliricidia-maize sites are more stable over time. Compared with the traditional fallow method, the yield of sorghum (*Sorghum bicolor* L. Moench) was found 55% higher in the two consecutive seasons after *Gliricidia* (Hall et al., 2006).

The high growth rate of plants and biomass production in the agroforestry system can provide a considerable soil C pool. The top 20 cm of soil in the tree-based agroforestry systems under the humid and subhumid tropical regions can sequester 25 and 1.6 Mg C ha⁻¹, respectively, which ultimately can reduce CO₂ emission (Mutuo et al., 2005). Albrecht and Kandji (2003) reported that agroforestry systems annually can sequester 95 Mg C ha⁻¹, and projected that over the next 50 years the terrestrial ecosystems could store 1.1–2.2 Pg C year⁻¹. Conversion of conventional cropland to agroforestry system can annually add about 3.1 Mg C ha⁻¹, while better management of an existing agroforestry system may further add 0.3 Mg C ha⁻¹ (Watson et al., 2000). It is projected that conversion of 630 Mha (million hectares) of global unproductive arable lands into agroforestry systems could sequester 0.58 Mt (million tons) C year⁻¹ in 2040 (Jose, 2009). Such systems possess high potentials to reduce CH₄ production (Baah-Acheamfour et al., 2016). Tree-based systems can offset CH₄ emissions to some

extent, while high-input traditional cropping systems may intensify CH₄ productions (Kwak et al., 2019). Forests are the major sinks of CH₄, but agricultural systems can reduce the sinking intensity by 50% or more (Mutuo et al., 2005).

20.4.3 Grassland system

Grassland system sequesters soil C by adding grass biomass and increases soil OM. Legumes provide huge prospects for a viable grassland-based production system. The system increases forage production, reduces/replaces N fertilizer because of BNF, mitigates and promotes adaptation to climate change (Luscher et al., 2014). In the grasslands, the range of annual N₂ fixed by legume-*Rhizobium* symbiosis is 100–380 kg N ha⁻¹ (Carlsson and Huss-Danell, 2003). The grass-legume system annually can transfer N from legumes to grass ranged from 10 to 75 kg ha⁻¹ (Pirhofer-Walzl et al., 2012). Therefore, the grassland-legume system can be recommended as a doable strategy to mitigate climate change.

20.4.4 Legume silvopastoral system

An agrosilvopastoral system refers to a system where perennial and herbaceous crops are grown simultaneously and have an integrated approach to livestock production. This practice is extensive in many areas of the world where weather uncertainty is particularly a threat to crop production. Cultivation of food or forage crops in between hedges of different legume trees like *ipil ipil* (*Leucaena leucocephala*) and gliricidia might be an effective approach to improve soil health, and crop yields, provide fodder, and supply fuelwood. The impact of climate change can partially be alleviated by intercropping of forages and legumes into the existing farming system. Supplementation of legumes can make the fibrous feedstuffs more digestible to animals, increases feed intake efficiency, and reduces CH₄ emissions from animal production (Luscher et al., 2016). An increase in feedstuff digestibility of ruminants reduces CH₄ emissions through intestinal fermentation by 15%–30%. (Verge et al., 2007; Gurian-Sherman, 2011). A 9-years study conducted in Brazil displayed that the inclusion of sweetheart (*Desmodium ovalifolium*) into a signal grass (*Brachiaria decumbens*) sward increased annual C sequestration in a 100 cm soil layer by 0.66–1.17 Mg ha⁻¹ (Tarre et al., 2001). It is demonstrated that legumes can increase OC contents in soils of pastures, while intensively managed permanent pastures and grasslands have the highest C sequestration potential (Soussana et al., 2004; Chan et al., 2011). Agrosilvopastoral and silvopastoral systems in Brazil were found effective in increasing C sequestration and reducing GHG emissions (Torres et al., 2017). Annual GHG emissions from agroforestry systems in Brazil were observed at 7.98 t CO₂ eq. ha⁻¹ (Torres et al., 2017), while from Indian rice field GHG emission was recorded 6332 t CO₂ eq. ha⁻¹ (Bhatia et al., 2012).

20.5 Role of legume crops in climate change adaptation in system-based approach

20.5.1 Legumes in cropping systems

Grain legumes are essential adjuncts to predominantly cereal-based cropping systems. They are well adapted to either summer or winter climatic conditions. These legumes are traditionally cultivated in diversified cropping systems of different countries of the world. Legume-based cropping systems are observed in different countries which are illustrated in Table 20.2.

Legumes in the cropping system can be incorporated as a vital candidate either into intercropping or into crop sequence. Intercropping of legumes with main crops has many advantages including higher yield, NUE, and water use efficiency (WUE) compared to monocropping (Inal et al., 2007). For example, in Bangladesh, common legumes intercropped to improve system productivity are broadcasted aman rice/grass pea, mustard + lentil, mustard + chickpea, broadcasted aman rice/pea, maize + cowpea, maize + soybean, sesame + mung bean, and maize + black gram. The efficiency of land use by legume-based intercropping may be determined by the land equivalent ratio and accordingly, species selection should be done properly to maximize production and economic return (Muoneke et al., 2007).

20.5.2 Variety selection under climate change

Legumes are the third-largest group of flowering plants but only a few species of the group are cultivated as crops. Important legumes are groundnut, pigeon pea, chickpea, soybean, hyacinth bean (*Lablab purpureus*), grass pea, lentil, lupine, horse gram (*Macrotyloma uniflorum*), lima bean (*Phaseolus lunatus*), kidney bean, pea, winged bean (*Psophocarpus tetragonolobus*), fava bean, black gram, mung bean, Bambara groundnut (*Vigna subterranean*), and cowpea (Allen, 2013; Nasim et al., 2017). The selection of improved varieties of legume crops is very important to boost their yields in a particular area under specific climatic conditions. Therefore, growing environment, namely, light, temperature, rainfall, humidity, and soil conditions must be taken into consideration to utilize the full genetically potential of the legume cultivars. Differences in the benefit of legumes to succeeding crops also vary due to varieties of the same species, soil, and environmental condition which must be taken into consideration during variety development focusing on future scenarios of climate change. In this regard, several authors reported that the yield of wheat increased by 30% in legume-wheat than wheat-wheat, and by 21% than wheat monocropping (Kirkegaard et al. 2008; Angus et al. 2015). Short-duration legumes are very useful for climate-smart agriculture especially in crop intensification, increased system productivity, and income

TABLE 20.2 Important legumes crops grown in different countries under various cropping systems.

Country	Cropping patterns	References
Bangladesh	Lentil (<i>Lens culinaris</i>)-jute (<i>Corchorus</i> spp.)-rice, lentil-maize-rice, lentil-fallow-rice, chickpea (<i>Cicer arietinum</i>)-fallow-rice, chickpea-rice-rice, grass pea-rice-rice, grass pea-jute-rice, cowpea (<i>Vigna unguiculata</i>)-fallow-rice, soybean-fallow-rice, soybean-jute-rice, groundnut (<i>Arachis hypogaea</i>)-fallow-black gram (<i>Vigna mungo</i>), fallow-mung bean-rice, mustard-mung bean-rice, black gram-jute-rice	Nasim et al. (2017)
China	Maize-soybean-wheat, soybean-wheat-potato, mung bean-maize-wheat, groundnut-maize-mung bean, maize-mung bean, maize-faba bean	Zeng et al. (2016)
India Western Himalayan region	Rice-chickpea/lentil/pea, maize-chickpea/pea, ragi (<i>Eleusine coracana</i>)-chickpea/lentil/field pea, maize-mung bean, pigeon pea-wheat, mung bean-mustard, common bean (<i>Phaseolus vulgaris</i>)-potato	Singh et al. (2005); Meena and Lal (2018)
Central Himalayan region	Rice-pea, sorghum-chickpea, sorghum + pigeon pea-fallow, pearl millet (<i>Pennisetum glaucum</i>) + pigeon pea-fallow, rice/maize-chickpea/lentil, mung bean-wheat, rice/maize-chickpea/lentil/pea, pearl millet-chickpea,	
Eastern Himalayan region	Rice- mung bean, rice-grass pea, maize-pigeon pea, maize-chickpea/lentil/pea, soybean/groundnut-rapeseed/mustard	
Indonesia	Soybean-rice-soybean, maize-rice-mung bean, rice-mung bean, rice-soybean-mung bean, rice-groundnut-mung bean, rice-rice-mung bean	Gowda et al. (2001)
Myanmar	Pigeon pea-sesame (<i>Sesamum indicum</i>), sesame-mung bean, rice-mung bean, rice-black gram, rice-chickpea, rice-cowpea, soybean-wheat, rice-soybean, sesame-groundnut, rice-groundnut	Gowda et al. (2001)
Nepal	Rice-lentil/cowpea/mung bean/mustard-maize/rice, wheat-lentil/cowpea/mung bean/mustard-maize/rice, rice-wheat/maize-cowpea/mung bean/mustard, rice- mustard/legumes-fallow, wheat- mustard/legumes-fallow	Khadka and Paudel (2010); Basnett et al. (2014)
Pakistan	Mung bean + groundnut-wheat, wheat-rice-pea/lentil, rice-grass pea-wheat, rice-wheat-cowpea, rice-wheat-mung bean, rice-wheat-black gram	Johansen et al. (2000)

(Haque and Rashid, 2016). They further mentioned that the cropping pattern rice-potato-mung bean-rice showed 74.43% higher productivity and 75.92% higher income over the existing rice-potato-rice cropping pattern. The selection of appropriate crops for adaptation of grain legumes to climate change in the intercropping system is dependent on growth patterns of crop plants, availability of light and production inputs (Seran and Brintha, 2010). Several intercropping systems, namely, mixed intercropping, row intercropping, and strip intercropping may be practiced depending upon the environmental condition. In general, when the main crop is tall statured then strip intercropping is better for enhancing system productivity. In some cases, paired row intercropping is suggested than normal row intercropping for better resource use efficiency. Intercropping systems maximize canopy coverage thus reduce soil temperature and minimizing soil moisture evaporation which is indicative of better adaptation of legume crops under climate change.

The yields of different legumes, namely, chickpea (1.04–1.11 t ha⁻¹), lentil (0.92–1.48 t ha⁻¹), and soybean (1.13–1.77 t ha⁻¹) in Australia, Bangladesh, and India are low (FAOSTAT, 2021). In China, yields of chickpea, groundnut, lentil, and soybean are found 5.23, 3.90, 2.40, and 1.87 t ha⁻¹, respectively, while in the United States are

1.73, 4.43, 1.40, and 3.19 t ha⁻¹, respectively (FAOSTAT, 2021). This indicates further scope for improvement in legume yields still exist through introducing high-yielding varieties and adopting best soil and crop management practices. Currently, significant improvement has been made by developing new varieties of legume crops in different countries of the world (Table 20.3). These varieties have high yield potential as well as other special features relating tolerant to pests, drought, high temperature, salinity, and other climate change variables.

20.5.3 Drought management

The requirement of water of a crop is an amount of water needed during its life cycle for consumptive use, optimum growth and ultimately providing the desired yield. Water requirement is, therefore, demand and the supply which is fulfilled by water supply through irrigation, seasonal rainfall and soil moisture (Reddi and Reddy, 1995). Legumes contributed to higher WUE and are capable of producing an acceptable amount of yields using less amount of water than that of other crops (Ghosh and Panja, 2016). The water requirement of legumes however depends on the growing season, soil conditions, crop species, amount

TABLE 20.3 List of legume varieties adapted to climate change.

Country	Crop	Variety	Tolerance	References
Bangladesh	Lentil	BARI masur 8	Blight disease	Chowdhury et al. (2020)
	Chickpea	BARI chola 10	Temperature	
		BARI chola 9	Drought	
		BU chola 1	Drought	
	Grass pea	BARI khesari 5	Salinity	
		BARI khesari 4	Drought	
	Cowpea	BARI felon 1	Salinity	
	Soybean	BU soybean 1	Salinity	
		BU soybean 2	Salinity	
India	Lentil	HUL-35	Drought	Singh (2001)
	Chickpea	ICCV93043	Drought	Bakhsh et al. (2003)
		ICCV 92944, K 850	Drought	Basu and Singh (2003)
	Pea	Acc. 623	Heat	Srikanthbabu et al. (2002)
Pakistan	Lentil	MI-30	Drought	Salam and Islam (1994)
Australia	Pea	WA 932	Drought	Al-Karaki and Ereifej (1999)
England	Pea	Solara	Drought	Grzesiak et al. (1997)
Syria	Lentil	FLIP 92-12L	Drought	Mohebodini et al. (2006)
		ILL 1861, ILL 784	Drought	Hamdi and Erskine (1996)
	Chickpea	ILC 1799	Drought	Sabaghpour et al. (2006)
		FLIP 87-59C	Drought	Singh et al. (1996)
	Pea	Local Aleppo 2	Drought	Al-Karaki (1999)

and distribution of rainfall. Generally, summer legumes like mung bean and black gram do not require irrigation as rain water is sufficient to grow the crops.

In the dry season, legumes meet their water requirement from conserved moisture in the soil profile. Under less stored moisture or lack of precipitation, legumes respond very well to supplemental irrigation. In such cases, one pre-sowing irrigation is enough for the successful cultivation of lentils, chickpea, and grass pea in Bangladesh (BARI, 2016). Cowpea is sown under residual soil moisture and it is effective if the crop is irrigated at preflowering stages to maximized cowpea seed yield. In the case of pea, supplemental irrigation is not necessary because of its very short duration varietal characteristics. Soybean and groundnut are grown under rainfed conditions although there is every possibility to reduce yield due to reduction in the supply of soil moisture under climate change. Increasing the soil moisture significantly increase yield and yield contributing parameters of soybean (Rajagopal et al., 1995; Kazi et al., 2002; Ruhul et al., 2009) that may be done either by supplementary irrigations or by preserving soil moisture by legume

residue as mulch. Legume mulch protects the soil from rain-drop erosion and increases soil water infiltration for future use. In the perspective of climate change adaptation mulch could be an alternate option to supplement irrigation for the successful cultivation of grain legumes (Bunna et al., 2011).

20.5.4 Soil management

The inclusion of legume crops in the existing cropping systems is a unique method for better soil management under the consideration of the anthropogenic forcing of the climate system. The cereal-based cropping systems initially have contributed significantly to increase crop production. But ultimately in the long run decreased the factor productivity of costly inputs and showed negative effects through ecological degradation and disintegration (Binder et al., 2010). Therefore, a combination of legumes with cereals is suggested to enhance soil functions through improving bio-physico-chemical properties for sustaining system productivity. Legumes can sustain soil productivity even under climate change by adding OM and nutrients, harboring soil

organisms, increasing SOC and cation exchange capacity (CEC), improving soil aggregation, and resistance to erosion. Cereal crops require extensive tillage practices which deteriorate soil physical properties. Contrary, legumes can be grown even at no-till (NT) or minimum tillage (MT) which improve aeration, moisture and nutrient holding capacity, infiltration, and soil structure (Burger and Kelting, 1999; Mousavi et al., 2009; Lithourgidis et al., 2011). Thus, incorporation of legumes in cropping systems can maintain sufficient soil moisture when there is every possibility of water scarcity due to high evapotranspiration as a result of high temperature of climate change variable. Some studies revealed that NT or MT significantly increased yields of legumes and exhibited positive impacts on SOC compared to TT (Christopher and Lal, 2007). However, such impact of legumes in rotation are more profound in the case of forage, green manures, and cover crops which contribute large amounts of OC and N to the soil which is very important to climate change mitigation (IPCC, 2007).

Soil chemical properties are also influenced by legume crops through regulating CEC, maintaining pH, mineralizing nutrients, and OC that are pertinent to climate change (Bajjiya et al., 2017). The quantity of nutrients available provided by legumes is dependent on residue retention and its decomposition rate, crop management, and soil types. It is expected that the decomposition of legume residue increases due to the thermodynamic effects of climate change. Decomposition and mineralization of legume residues supply available nutrients in soils for the standing and following crops (Melero et al., 2007). Microbial biomass is a key parameter for assessing soil biological health and legume crops provide a congenial environment to increase microbial diversity and quantity in soil (Deakin and Broughton, 2009).

20.6 Role of legumes in climate change mitigation in system-based approach

Leguminous plants could play a vital role by providing various services in conjunction with climate change mitigation principles. Cropping systems that include legume crops might be considered as an effective tool for soil C sequestration which could play important role in climate change mitigation strategy (Meena et al., 2015c).

20.6.1 Legumes reduce external nitrogen fertilizer use

Atmospheric N fixation by the legume crop could be an option to reduce the demand for industrial N fixation. Nitrogen content in the atmosphere is 78% which is inert and unavailable for plants to utilize. But legumes with a microbiological process, known as BNF convert the atmospheric N₂ into a plant utilizable form. Among different N

fixing systems, legume-*Rhizobium* symbiotic N fixation is the most valuable system. Meena and Lal (2018) reported that globally the amount of BNF is 250 Tg N year⁻¹ of which 110 Tg from the land ecosystem and the remaining 140 Tg under the marine system, which is higher than that of Haber-Bosch N (Meena and Lal, 2018). Legume-*Rhizobium* system ensures economically reasonable and environmentally sustainable ways of lowering external N fertilizers derived mainly from industries through Haber-Bosch techniques.

In addition to the significance of BNF in standing leguminous plants, it could also decrease the demand of N for succeeding nonlegume crops. Grain legumes may save nearly 170–220 kg N ha⁻¹ compare to nonlegumes (Varma et al., 2017). Furthermore, mineralization of root nodules and legume residues provides different nutrients including N, and thus reduces demand for N fertilizer for the following crops (Crews and Peoples, 2004; Pikul et al., 2008). Annual N fixation of different legume species through BNF vary greatly where 465 kg N ha⁻¹ was fixed by alfalfa (*Medicago sativa* L.), and pinto peanuts (*Arachis pintoi*) as green manure contributed 14 kg N ha⁻¹ (Table 20.4). Most of the legume crops could satisfy their N requirement from the BNF. Results presented in Table 20.4 indicated that annual N fixations from legume-*Rhizobium* symbiosis were 101 and 201 kg ha⁻¹ in groundnut and cowpea, respectively, that could satisfy about 79, and 89% of their N requirements, respectively (Dakora et al., 1987). There are 19320 species of legume plants across the world and they vary in sizes, life spans, root systems, etc., and accordingly their N fixation capacity also vary. However, it has been reported from a case study conducted in different regions of Europe that up to 20 kg of N fertilizer could be saved from 1 ha of land of faba bean, lupines, pea, and forage legume in succeeding wheat (Bues et al., 2013). Thus, biologically fixed atmospheric N₂ by legume crops would essentially reduce the use of external N fertilizer use. Similar to N, legumes have a positive effect on P availability in soil and this reduces the application of P fertilizer. The roots of legumes release exudates and organic acids which enhance P availability (Meena et al., 2017).

The increasing demand for N fertilizer in agriculture would require considerable energy consumption for fertilizer manufacturing and application. In this context, legume-based farming practices in agriculture and the agro-forestry system could serve a colossal role in environmental sustainability and economic viability through BNF. It has been reported that total energy consumed by the legume cropping system (pea) was 55% lower as compared to the nonlegume crop barley in Denmark (Peoples et al., 2009). In the same study, it was further documented that total fossil energy used by the grass-clover mixture was 41% lesser than that of the sole pure grass systems. A similar trend was evident in different locations. European grain legumes could reduce up to

TABLE 20.4 Nitrogen fixation by different legumes.

Legume	Annual N from BNF (kg N ha ⁻¹)	References
Groundnut	101	Dakora et al. (1987)
Cowpea	201	Dakora et al. (1987)
Alfalfa	465	Anglade et al. (2015)
Red clover (<i>Trifolium pratense</i> L.)	252	Anglade et al. (2015)
White clover (<i>Trifolium repens</i> L.)	102	Anglade et al. (2015)
Faba bean	165	Anglade et al. (2015)
Field pea	111	Anglade et al. (2015)
Lentil	52	Anglade et al. (2015)
Velvet bean (<i>Mucuna pruriens</i>)	59.6	Okito et al. (2004)
Pinto peanuts as green manure	14	Mendonça et al. (2017)
Pigeon pea as green manure	44	Mendonça et al. (2017)
Lablab beans (<i>Dolichos lablab</i>) as green manure	33	Mendonça et al. (2017)
Pea	90–128	Jensen (1996)
Chickpea	64–103	Fatima et al. (2008)
Mung bean	19–54	Hayat et al. (2008)
Black gram	16–79	Hayat et al. (2008)

35% energy consumption than that of other crops (Reckling et al., 2014). Under Swiss conditions, faba bean and field pea required 39%, 31%, and 60% less energy when compared with wheat, barley, and maize, respectively (Kopke and Nemecek, 2010). Whereas in the North American situation, energy requirement in field pea was 23%, 45%, 47%, 35%, and 53% lower than soybean, spring wheat, winter wheat, flax, and maize, respectively (Rathke et al., 2007). The energy-saving by the legume crops is largely associated with the less demand for industrial N that require a significant amount of energy during its production through Haber–Bosch process. Moreover, a substantial amount of energy could be saved from related transportation, storage, and field application of N fertilizer. Thus, the inclusion of legume crops in the cropping systems would bring multiple benefits including energy savings in agriculture.

Under the European agriculture system, total N inputs by fertilizer and BNF varied from 56 to 174 kg ha⁻¹ for legume crops, whereas average N fixation by the legumes varied from 32 to 115 kg ha⁻¹. In contrast, the usual mineral N input for spring cereal is around 100 kg ha⁻¹ and for oilseed or winter cereal is 200 kg ha⁻¹ (Fertiliser Practice, 2014). For the Indo-Gangetic Plains of India, Sapkota et al. (2020) suggested 120–200 kg N ha⁻¹ and 50–185 kg N ha⁻¹ for rice and wheat production, respectively. In comparison, it has been reported that a very small dose of N (15–25 kg ha⁻¹) as

a starter dose is enough to meet the N requirement for most of the pulses grown in India can meet 80%–90% of their N requirement through BNF (Nadarajan et al., 2013). It has been estimated that including transportation and storage it requires around 65 to 100 MJ energy for the production of per kg N fertilizer (Wood and Cowie, 2004; Kongshaug, 1998) depending on types of factories and fossil fuel. Fossil fuel-based N fertilizer production consequences 2.1–5.5 kg CO₂ eq. GHG emission from 1 kg of N fertilizer (Jensen et al., 2020). As the legumes save a substantial amount of N fertilizers as compared to the cereal crops, ultimately GHGs emissions will get reduced considerably.

20.6.2 Legumes reduce water use and associated energy

To keep the agricultural system alive and viable the use of production inputs needs to be reduced in the face of anomalous drought and other extreme events of climate change (Brand and Jax, 2007). Climate change has seriously affected the ground and surface water cycles by regulating the seasonal changes of precipitation patterns throughout the world. The excess utilization of groundwater for the irrigation of cereal crops has made the situation worst in climate-vulnerable countries especially in South Asia including India and Bangladesh. Less water requirements, and ability

to resist drought qualify legumes as highly efficient and potential means of addressing the negative effect of climate change arises from cereal-based cropping systems (Kumar et al., 2018a). Grain legumes need about 250–300 mm of water, while rice, wheat, and sugarcane require 900–2500, 400–450, and 1400–2500 mm, respectively (Kumar et al., 2018a). They also observed that production of chickpea, wheat, and rice from 1 ha-mm of water were 12.5, 7.0, and 2.5 kg, respectively. Under Mediterranean water-limited conditions, Pala et al. (2007) confirmed the comparatively higher WUE in wheat-lentil rotation followed by wheat-chickpea and wheat-wheat system. Most of the pulse crops have a shallow root system and utilize water only from the top few cm of the soil layer, while leave water in the deeper layer for the subsequent cereals and or oil crops.

In irrigated crop fields, one of the main causes of GHG emissions is the use of energy for operating irrigation pumps (El-Gafy and El-Bably, 2016). Less water demand of legumes as compared to the cereals would ensure less diesel and electricity consumption for water-lifting devices. Less energy consumption will thus reduce GHGs emissions and ensure climate change mitigation. It has been demonstrated that lifting of 1 m³ of water for irrigating the crop field in Egypt by nonimproved and improved diesel pumps emitted 1.11 and 0.87 kg CO₂, respectively (El-Gafy and El-Bably, 2016). The CO₂ emission from 1 ha of crop field irrigated by improved and nonimproved diesel pumps was 5.95 and 7.16 t, respectively. In their report, they also showed that when irrigated with diesel pumps at the Hamed Minisy canal of Egypt, the mean CO₂ emission per ha of broad bean was much lesser (3.32 t ha⁻¹) as compared to rice (15.23 t ha⁻¹), wheat (4.13 t ha⁻¹) and maize (6.97 t ha⁻¹), respectively.

20.6.3 Legumes require less demand of tillage and other machineries

Legume crops require less tillage for cultivation. Different legumes, namely, lentil, grass pea, mung bean, etc., can be accommodated in rice fields under relay cropping (Gupta and Kumar, 2019). Kumar et al. (2018a) found that CT increased the grain yield of chickpea compared to TT, and the resultant yield increment was 28.2% in NT dibbling + mulching, and 22.7% in NT drill + mulching system. Blackshaw et al. (2007) conveyed that dry bean can be grown successfully under NT systems in western Canada, which is a very common practices for lentil and field peas in this region. They further confirmed that plant establishment, pest management, maturity date, as well as yield of dry bean were not negatively affected in an NT system as compared to the TT. Cultivation of lentils under rainfed conditions following wet season rice with CT confirmed a higher yield of lentils in the eastern part of India (Bandyopadhyay et al., 2016). The practice is furthermore suitable for reducing

cultivation costs through the reduction of labor and farm power (Kumar et al., 2014; Meena et al., 2015a).

Legumes like lentil, chickpea, pea, faba bean, groundnut, and soybean play a leading role in CT in many countries including Australia, Canada, Brazil, the United States, and Turkey. Compared to TT, RT, and NT systems showed significant positive effects on SOC status (Christopher and Lal, 2007). Under TT practices, different types of heavy farm machineries including moldboard plough and disk-harrow are used to prepare fields making suitable for the cultivation of legumes. In contrast, RT or NT under conservation agriculture requires less utilization of farm machineries that will save related energy and labor cost without compromising yields of legumes.

Legume can be cultivated either adopting NT or MT which potentially save energy from tillage operation through agricultural machineries and also save time and energy from ploughing using draft animals (Reckling et al., 2014). Maraseni and Cockfield (2011) reported less GHG emissions from the NT system when compared with TT practices in Australia. The lesser emissions are directly related to reducing passes of farm machinery in the field for tillage which ultimately ensures less diesel consumption and it will pay to a reduction in GHG emission. In their experiment, Maraseni and Cockfield (2011) showed that fuel and machinery related emissions under NT are nearly half as compared to the TT. They further reported that when switched from TT to NT, reduction of total GHGs emission was highest in legume crop chickpea (25%) and lowest in cereal crop wheat (5.4%). From a long-term field experiment conducted by Utomo et al. (2012), it was observed that CO₂ emission from CT practice was continuously lower throughout the season as compared to the TT. The higher CO₂ emission from TT might also be due to the enhanced microbial activity as tillage operation inverted the soil to permit O₂ entry and mixed with the residues and organic particles which ultimately release more CO₂ (Rastogi et al., 2002).

20.6.4 Legumes for better carbon sequestration in soil

Legumes are important in terms of C capture, while enhanced C accumulation is greatly related to reduced GHG emissions. The inclusion of legumes in crop agriculture is a dynamic choice for C sequestration in soil. Benefits of the addition of legumes in rice rotation in terms of C sequestration, BNF, N mineralization, and microbial diversity are depicted in Fig. 20.5. The ability of soil C sequestration varies greatly among different legumes based on biomass production and residue retention in soil. In addition to aboveground biomass, higher belowground biomass producing ability and leaf-shedding nature of the legumes ensure C sequestration greatly compared to the nonlegume cereals and other crops (Ganeshamurthy, 2009). The

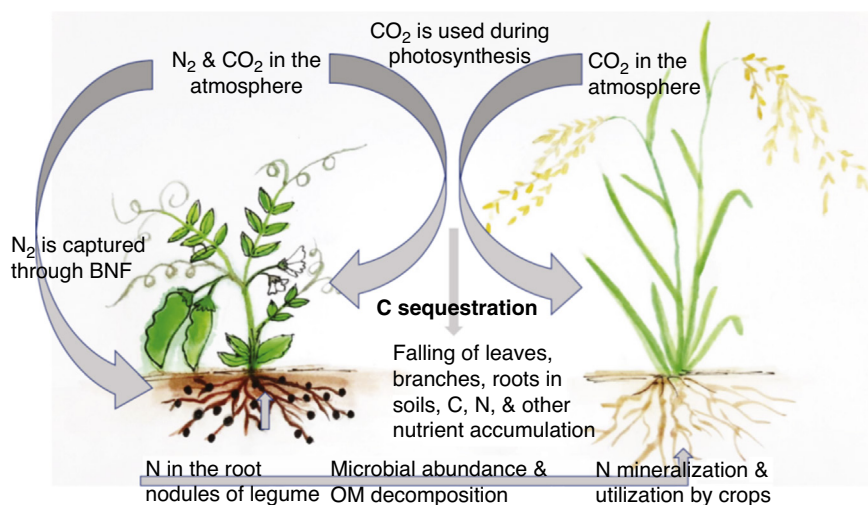


FIG. 20.5 Biological nitrogen fixation, microbial abundance, C sequestration, and N availability in soils of legume-rice cropping system (Courtesy Mahita Rahman Noushin, Dhaka, Bangladesh).

TABLE 20.5 Increase of soil organic carbon stock from legumes/legume-based cropping systems

Legumes/legume-based cropping system	Study period (years)	Increase of SOC stock (Mg C ha ⁻¹)	References
Alfalfa	7	24.1	Guan et al. (2016)
Bush clover (<i>Lespedeza davurica</i> S.)	7	19.9	Guan et al. (2016)
Milk vetch (<i>Astragalus adsurgens</i> Pall.)	7	14.6	Guan et al. (2016)
Alfalfa pasture	10	16	Zhang et al. (2009)
Maize-soybean intercrop	2	18	Chidowe et al. (2017)
Maize-Desmodium intercrop	2	8.82	Chidowe et al. (2017)
Maize strip cropped with soybean (2:4)	2	12	Chidowe et al. (2017)
Maize strip cropped with Desmodium (2:4)	2	28.35	Chidowe et al. (2017)
Pinto peanuts	6	44.7	Fisher et al. (1994)

legume-based cropping system contributed to the increased SOC stock which is endorsed by many scientific studies (Table 20.5). The greater ability of legumes for N fixation through BNF also contributes to better C sequestration (0.88 Mg C ha⁻¹ year⁻¹), while the grasses sequester 0.71 Mg C ha⁻¹ year⁻¹ (Diekow et al., 2005; Martin et al., 2012).

Guan et al. (2016) reported that when arable land changed to perennial pasture using grass legumes significantly improve SOC stock in China. The SOC stocks were amplified by 14.6, 19.9, and 24.1 Mg C ha⁻¹ in milk vetch (*Astragalus spp.*), bush clover (*Lespedeza thunbergii*), and alfalfa-oriented pastures, correspondingly (Table 20.5). Furthermore, they stated that C stock was declined by 4.2 Mg C ha⁻¹ in the bare fallow land. From a 10-year study of alfalfa pasture in northwest China, an increment of SOC by 16 Mg C ha⁻¹ in the topsoil (0–30 cm) has been achieved (Zhang et al., 2009). Legume pastures receiving adequate precipitation and

fertilizer could sequester a higher amount of soil C as compared to the legumes grown in unfertilized and unirrigated conditions in a semiarid environment (Guan et al., 2016). It has been reported that a starter supply of 20 kg N ha⁻¹ for the cultivation of legumes, namely, cowpea, soybean, and cavalcade centro (*Centrosema pascuorum*) improves C status in soil (Adeboye et al., 2006). The vigorous growth of the legume plants due to the BNF could assimilate a higher amount of atmospheric CO₂ through photosynthesis. Upon incorporation and subsequent decomposition of the legume biomass in the soil would return this assimilated C in soil. In Argentina, soybean-maize intercropping (3:2) ensured buildup of 23.6 g C kg⁻¹ soil, while sole maize contributed 21.8 g C kg⁻¹ soil (Bichel et al., 2016). Under the Asian situation, the inclusion of green gram in the rice-wheat cropping pattern after wheat harvesting yielded dry matter of 4.5 Mg ha⁻¹ which might increase soil C (Meena et al., 2015a, b, c).

20.7 Ways to remove barriers in the way of inclusion of legumes in cropping system

There are many barriers associated with different crop cultivation under varied ecological conditions that warrant a range of solutions to include legumes in the existing production environment. In a nutshell, the following points need to be considered to remove barriers for improving pulses production around the globe.

- Adequate budget and supports from the public and private sectors for growing pulse crops.
- Benefits of cereal-pulse rotations on total land productivity need to be disseminated through participatory field trials.
- Production of pulses is hindered because it needs to compete with rice, wheat, maize, potatoes for space along with socioeconomic and cultural constraints. So, location or region-specific efforts are needed to alleviate bottlenecks in improving pulse production.
- Many farmers are not aware of higher N contents in legume residues than cereals that improve rumen digestion and animal growth. Such benefits need to be demonstrated among legume growers.
- Necessary steps should be taken in linking markets, in aligning to market, and in fulfilling consumers' demands for the successful entry of pulses in the profitable farming system. Therefore, improvement in research-extension-farmer linkages and establishment of agro-processing industries are needed.
 - Other steps that can be considered for alleviating constraints on pulse production could be: Encouraging smallholder farmers to grow pulses as intercrop in terrace and *jhum* culture with maize; pulse cropping with rice, wheat, maize, sugarcane, etc.
 - Growing pigeon pea, mung bean, black gram, etc. on the bunds.
 - Legumes as a component agroforestry system should be considered.

20.8 Research priorities and resource development of climate-smart legume-based system

Demand for pulse crops and their growing environments are very much different around the globe. So, research priorities and resources development will vary greatly. The following points can be considered:

- Generation of new legume crops tolerant to insects, diseases, heat, waterlogging, drought, salinity, acidity, etc., along with the development of appropriate agronomic management practices and their dissemination among the growers.

- Improvement of smallholder farming systems based on local demands in hilly regions.
- Integrated nutrient management for legume-based cropping systems.
- Generation of technology for increased use of legumes in dietary contributions to human health and animal productivity.
- Emission scenarios for different pulse crops need to be developed for minimizing GHG emissions.
- Determination of ecosystem service potentials of different pulse crops.
- Investigations on pests and disease cycle breaking potentials of legumes under varied ecology of cereal production are important avenues.
- Management options for growing legumes after rice in tropical Asia should be found out.
- Seed rate, its placement and fertilizer management options for growing legumes in association with rice, wheat, and maize crops.

20.9 Possible future policy and action plan

The agri-food system that includes farmers, businesses and markets, regulation policies and incentives, infrastructural development, education, and research policies need improvement. Some of the points could be:

- Reduction of production costs and risks for the profitability of grain legumes cultivation.
- Increased utilization of legume products for expanding marketing opportunities.
- Improvement of value chains for legume marketing.
- Capacity enhancement of agricultural research organizations in developing countries for the improvement of pulses.
- Establishing a balance among producers, processors and consumers through policy intervention.
- Strategy for variety development should meet the demands of growers and market vis-à-vis consumers.
- Community-based interventions are needed to improve the adoption and yield of pulse crops. Food-feed legume crops for dual purposes are time demanding for increased farm productivity through the inclusion of livestock.
- National policies should give thrust to nutritious food availability, but not to carbohydrate production only.
- The use of legume flours instead of cereals in extruded foods could be a viable strategy to increase the consumption of pulses for improving nutrition security in society.
- Enhancing crop rotations or as inter- and mixed-cropping with legumes can help in reducing the impacts of deforestation under changing climate.
- Promoting the use of legume residues as an inexpensive fodder in cattle production will increase the farm income of smallholder farmers.

- P availability from normally unavailable form become available because of legume cultivation- the mechanism needs to be investigated fully.
- Policy support is needed for crop diversification that can improve WUE and expansion of food baskets.
- Policy support and emphasis are needed in proving machinery for sowing and harvesting.
- About 60% of the global poor live in dryer parts of Africa and Asia. Improving dryland cereal and pulse production can emancipate the burden from hunger.
- Colearning among scientists, advisors, and farmers for designing new cropping systems through the inclusion of pulses and for the generation of best agronomic management practices are needed to meet future challenges of food production.
- Addition of food grain legumes in the procurement policies of the government. Such practice will help in achieving nutrition security.
- Price incentives in terms of fertilizer and irrigation water for the inclusion of leguminous crops with nonlegume crops cultivation as sequential or intercrop would be beneficial.

20.10 Conclusions

Legume-based cropping systems may immensely contribute to foodstuff and nutrition security for the ever-burgeoning population of the world. Such cropping systems not only increase yields of legumes but also nonlegumes including cereals and vegetables. Traditional cropping systems with higher N inputs contribute to GHG emission and subsequent global warming and climate change. The inclusion of legume crops in the existing cereal-based field crops and agroforestry systems might ensure food, feed, and fiber production, and environmental sustainability. Legume-based cropping systems ease the crop production systems and bring multiple benefits. For example, no-till and minimum tillage can be implemented for legume cultivation which might save energy from tillage operation. Adoption of no-till for chickpea cultivation could reduce about 25% GHG emissions. Climate vulnerability is a great concern for agricultural sustainability and makes food production and distribution unsafe in many countries. Legume crops can be grown in harsh environments like drought and saline conditions. Therefore, legume-based systems can ensure ecological integrity, climatic adaptability, and safeguard and increase crop production. Cultivation of pulses and cereals in rotations or the association may increase the yield of cereal by about 25% than only cereal. The yields of different legume crops are still low enough and widely varied across the world. Hence, the development of high yielding, drought, salinity and temperature tolerant legume varieties are needed to be accompanied by nonleguminous oil and cereal crops. Legume has diverse potentials to make

the production systems viable through soil health improvement. Legumes play a key role to increase nutrients including N, sequester C in soils, harbor soil microbes, ensure conservation agriculture, and establish a congenial environment for BNF. Annually, alfalfa alone and maize-soybean intercrop sequester 3.44 and 9.0 Mg C ha⁻¹, respectively. Legumes and legume-based cropping systems may annually fix roughly 50–200 kg N ha⁻¹ through BNF. Because of BNF, inorganic N fertilizer application in pulse crops can be reduced by 80%–90%. Therefore, the obvious resultant effect is the reduction of GHG emissions from N fertilizer application. Terrestrial BNF exceeds the amount of reactive N produced through the Haber–Bosch process which is a noble contribution of legumes for mankind. Different barriers in relation to socioeconomic, climatic, and agronomic management concerns for legume cultivation need to be judiciously addressed. Appropriate future research avenues on legume-based cropping systems should be launched considering carbon and nitrogen footprints, ecological veracity, and varietal development adaptive to a harsh environment. There is still ample scope remaining to advocate and adopt legume-based cropping systems around the world where policy recommendations might act to remove associated barriers in this regard.

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Abbreviations

BNF	Biological nitrogen fixation
BARI	Bangladesh Agricultural Research Institute
BMPs	Best management practices
C	Carbon
Ca	Calcium
CEC	Cation exchange capacity
CH ₄	Methane
CO ₂	Carbon dioxide
CO ₂ eq.	Carbon dioxide equivalent
CT	Conservation tillage
FAOSTAT	Food and Agriculture Organization Corporate Statistical Database
Gg	Giga gram
GHGs	Greenhouse gases
GWP	Global warming potential
IPCC	Intergovernmental Panel on Climate Change

K	Potassium
Mha	Million hectare
Mg	Megagram
MT	Minimum tillage
N	Nitrogen
NH ₃	Ammonia
NO	Nitric oxide
N ₂ O	Nitrous oxide
NO ₃ ⁻	Nitrate
NT	No-till
NUE	Nutrient use efficiency
O ₂	Oxygen
OC	Organic carbon
OM	Organic matter
P	Phosphorus
ppm	Parts per million
RT	Reduced tillage
S	Sulfur
SDGs	Sustainable Development Goals
Tg	Tera gram
TT	Traditional tillage
WUE	Water use efficiency
SOC	Soil organic carbon
UN	United Nations

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